CDS 101 -Final Project Report

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1. Problem Definition

- Our project investigates whether weather conditions are related to equine incidents at New York race tracks. The main question we are studying is: How significant is the relationship between equine incidents and weather conditions?
- This topic is important because equine deaths and injuries are serious safety concerns in horse racing. By studying incident patterns over time and comparing them with weather conditions, we may be able to identify whether certain weather factors are associated with higher or lower incident counts.
- The objective of this analysis is to explore trends in equine incidents from 2009 to 2025, compare incidents across different years, tracks, incident types, seasons, and weather categories, and determine whether weather appears to have a noticeable relationship with incident patterns.
- A key assumption is that the incident reports are accurate and consistently recorded. Another assumption is that the weather information connected to each incident is reliable enough to be grouped into categories such as clear, rainy, cloudy, snowy, or unknown.

2. Data Acquisition & Description

- The name of the dataset is Equine Death and Breakdown from the data.gov website which is an official website of the United States government.
- Obtained through a download link from data.gov website.
- There are a lot of variables / features such as Incident Type, Horse, Jockey Driver, Track, and so forth but we will be focusing primarily on two variables, Death or Injury and Weather Conditions. Death or Injury variable shows if a horse died in an incident or was just injured. Weather Conditions variable shows the weather conditions at the time of the incident. We will also be using “Track”, “Year”, and “Incident Date”, and “Incident Type” columns to help supplement our findings.
- The dataset has 5,383 rows and 13 columns.

3. Data Cleaning & Preprocessing

- This dataset was extremely messy and we had to do a lot of cleaning to make it usable and went from 5,383 rows to just 1,702. We got rid of any rows that had NA values in the Weather Condition and Death and Injury variables. The incident date variable had to be changed to a date type. The column names were fixed to make it easier to use and read. Originally the dataset's weather condition variable had 1000+ different unique entries but we wanted them to be categorized in just a few simple ones. The weather_clean variable that was created that holds the weather_conditions values without all the unnecessary information and spaces. This is then piped into weather_group which categorizes the specific weather condition happening at the time of the incident such as "Rain", "Snow", or "Clear" and so forth. The death_or_injury_clean variable cleans up the 10 unique values and categorizes them into 2 simple ones "death" or "injury." After creating these new variables we filtered them to only include the values that were relevant to our research question. A season variable was also created for EDA. Finally, the Death and Injury variable was turned to binary 0/1 variable since its a categorical variable and we need a numerical values to make a linear regression model.

```r
#Fix column names
horses_clean <- horses %>%
  rename(year = Year, incident_date = `Incident Date`, track = Track,
         inv_location = `Inv Location`,
         racing_type_description = `Racing Type Description`,
         division = Division, weather_conditions = `Weather Conditions`,
         horse = Horse, trainer = Trainer, jockey_driver = `Jockey Driver`,
         incident_description = `Incident Description`,
         death_or_injury = `Death or Injury`,
         incident_type = `Incident Type`)

#Fix dates
horses_clean$incident_date <- mdy(horses_clean$incident_date)

#Get rid of NA entries
horses_clean <- horses_clean %>%
  filter(!is.na(weather_conditions), !is.na(death_or_injury))

#Fix weather_condition column
#Used ChatGPT to help with this cleaning part
horses_clean <- horses_clean %>%
  mutate(
    weather_clean = weather_conditions %>%
      str_to_lower() %>%
      str_trim() %>%
      str_replace_all("[0-9]+", "") %>%
      str_replace_all("[^a-z ]", "") %>%
      str_squish()
  ) %>%
```

```r
  mutate(
    weather_group = case_when(
      str_detect(weather_clean, "rain|shower|drizzle|raine") ~ "Rain",
      str_detect(weather_clean, "snow|flurr|sleet") ~ "Snow",
      str_detect(weather_clean, "thunder|storm") ~ "Storm",
      str_detect(weather_clean, "cloud|coudy|cloudy|clody") ~ "Cloudy",
      str_detect(weather_clean, "clear|sun|fair|dry") ~ "Clear",
      str_detect(weather_clean, "windy|fog|mist|hazy|wind") ~ "Other Weather",
      weather_clean %in% c("", "na", "n a", "unc") | is.na(weather_clean) ~ "Unknown",
      TRUE ~ "Other"
    )
  )

#Clean death_or_injury column
horses_clean <- horses_clean %>%
  mutate(
    death_or_injury_clean = death_or_injury %>%
      str_to_lower() %>%
      str_trim(),

    death_or_injury_clean = case_when(
      str_detect(death_or_injury_clean, "death|dead|euthanasia") ~ "Death",
      str_detect(death_or_injury_clean, "lame|injury|lameness") ~ "Injury",
      TRUE ~ "Other"
    )
  )

horses_clean <- horses_clean %>%
  filter(death_or_injury_clean != "Other") %>%
  select(year, incident_date,
         incident_type,
         track,
         incident_description,
         weather_group,
         death_or_injury_clean) %>%
  filter(weather_group != "Other" & weather_group != "Unknown")

#Create season variable (Code reference linked below)
horses_clean <- horses_clean %>%
  mutate(
    season = case_when(
    month(incident_date) %in% 9:11 ~ "Fall",
    month(incident_date) %in% c(12,1,2) ~ "Winter",
    month(incident_date) %in% 3:5 ~ "Spring",
    TRUE ~ "Summer"
  ))
```

4. Exploratory Data Analysis (EDA)

```
summary(horses_clean)
```

```
##       year      incident_date        incident_type          track
##  Min.   :2009   Min.   :2009-04-28   Length:1702        Length:1702
##  1st Qu.:2012   1st Qu.:2012-01-20   Class :character   Class :character
##  Median :2016   Median :2016-10-01   Mode  :character   Mode  :character
##  Mean   :2017   Mean   :2017-03-27
##  3rd Qu.:2022   3rd Qu.:2022-09-15
##  Max.   :2026   Max.   :2026-04-12
##  incident_description weather_group      death_or_injury_clean
##  Length:1702          Length:1702        Length:1702
##  Class :character     Class :character   Class :character
##  Mode  :character     Mode  :character   Mode  :character
##
##
##
##     season
##  Length:1702
##  Class :character
##  Mode  :character
##
##
##
```

i.

```
library(ggplot2)
ggplot(horses, aes(x = Year, color = 'white')) +
geom_line(stat = "count") +
geom_point(stat = "count", size = 3) +
labs(
  x = "Year",
  y = "# of Incidents",
  title = "Equine Incidents Over Time"
  ) +
theme_minimal()
```

- The data overall shows a clear decline in equine incidents from 2009 to 2026 with few inconsistencies. Incident counts start at a moderate level in the late 2000s and rise very fast and peak in 2012. After that the count steadily falls. This pattern equine incidents have generally decreased over the years. The inconsistencies show how the incident counts do not decrease perfectly smoothly, indicating that many external factors such as heat, cold, rain, etc. contribute to short term increases.

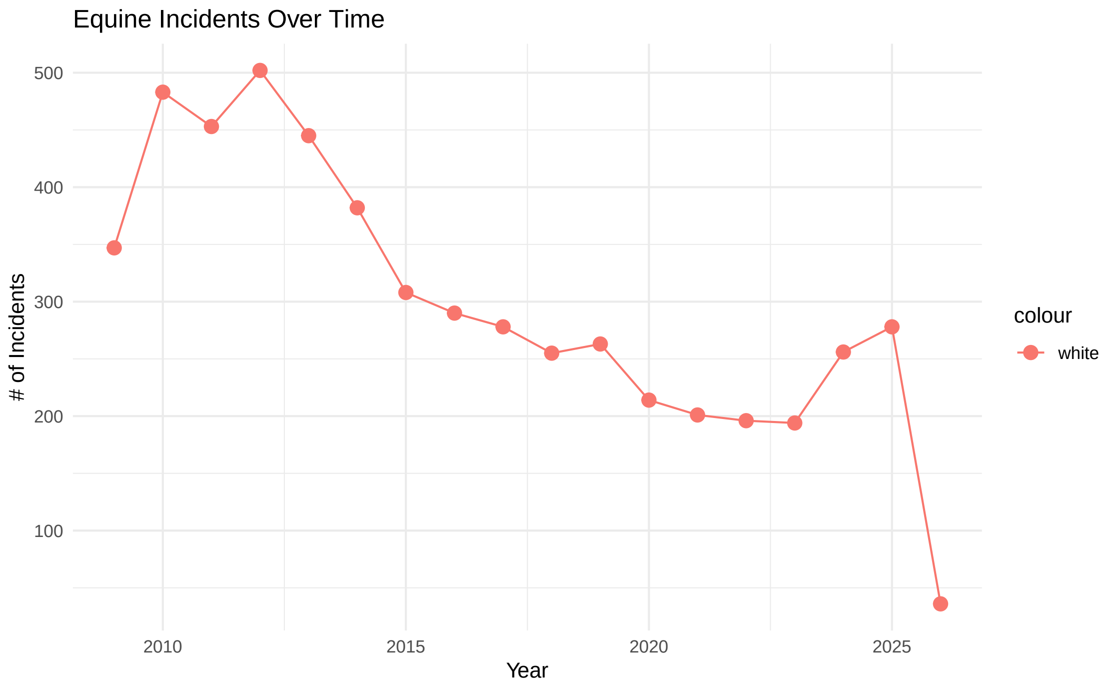
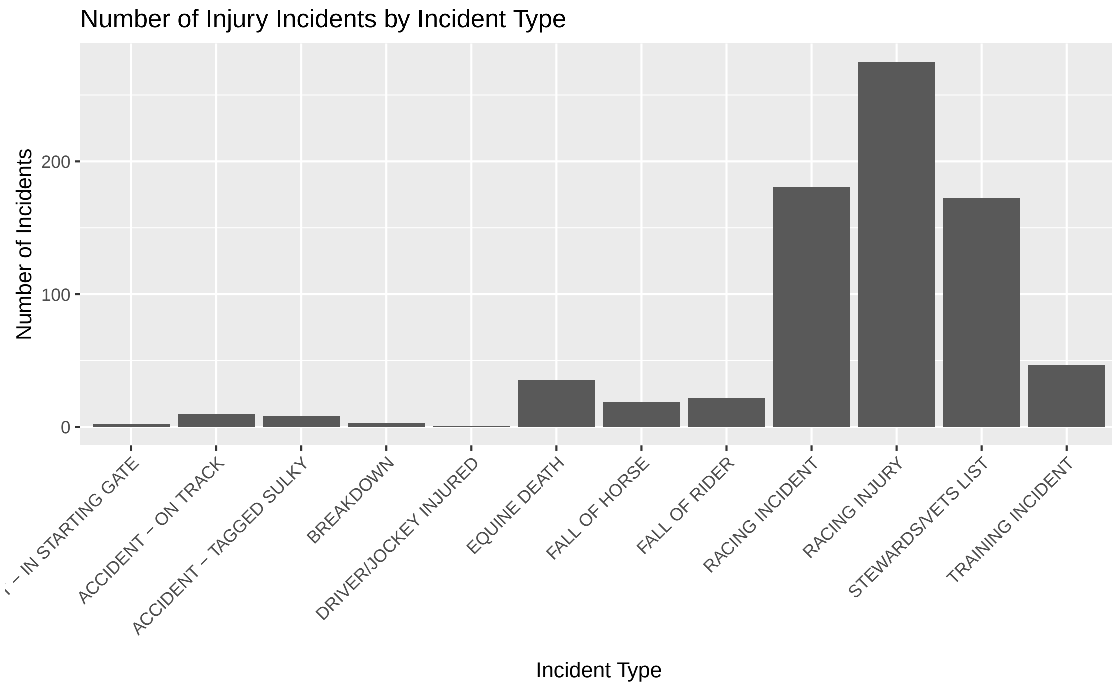


Figure 1: Relationship between time of year and weather related equine incidents

ii.

```r
horses_clean %>%
  filter(death_or_injury_clean == "Injury") %>%
  ggplot(aes(x = incident_type)) +
  geom_bar() +
  labs(
    title = "Number of Injury Incidents by Incident Type",
    x = "Incident Type",
    y = "Number of Incidents"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



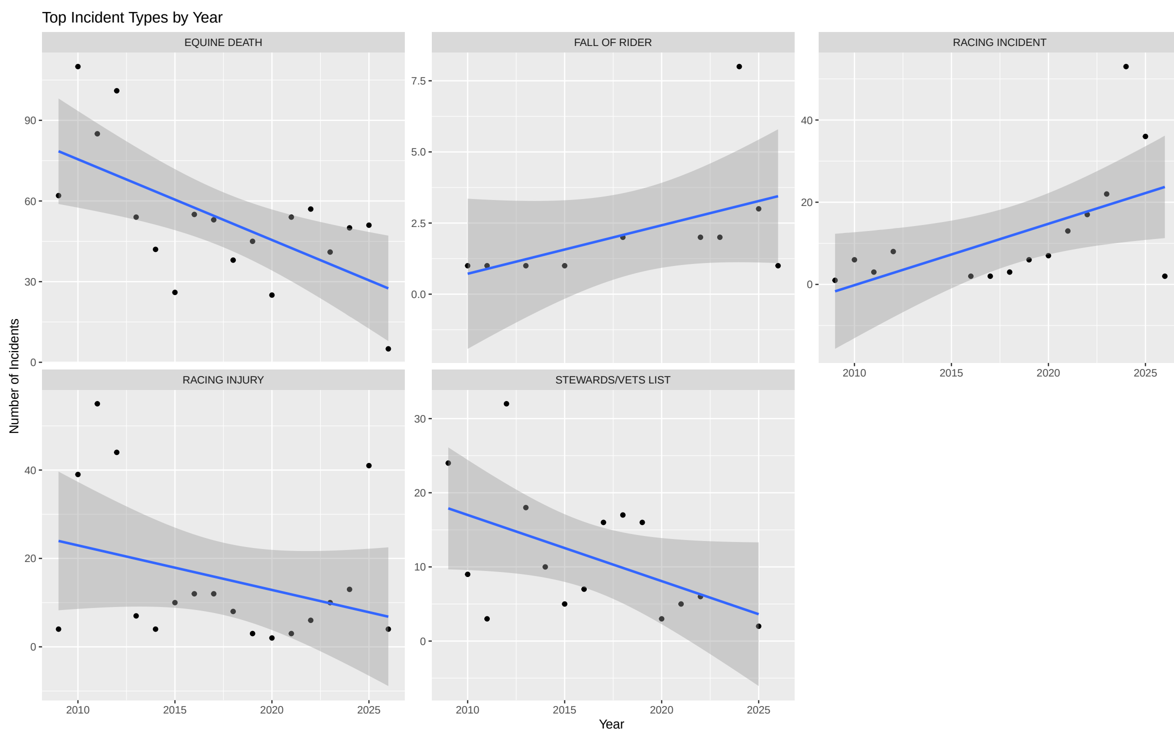
- This graph connects to our research question by showing which types of equine incidents happen most often in the data set. Since our project is studying the relationship between weather related variables and equine deaths or breakdowns, we first need to understand what kinds of incidents are being recorded. The most common incident types are the main categories we can later compare with weather conditions, temperature, precipitation, or season. This graph does not show a weather relationship by itself, but it gives important background for the weather analysis.

iii.

```
horses_clean %>%
  filter(incident_type %in% c(
    "EQUINE DEATH",
    "STEWARDS/VETS LIST",
    "RACING INCIDENT",
    "RACING INJURY",
    "FALL OF RIDER"
  )) %>%
  count(year, incident_type) %>%
  ggplot(aes(x = year, y = n)) +
  geom_point() +
  geom_smooth(method = "lm") +
  facet_wrap(~ incident_type, scales = "free_y") +
  labs(
    title = "Top Incident Types by Year",
    x = "Year",
```

```
    y = "Number of Incidents"
  )
```

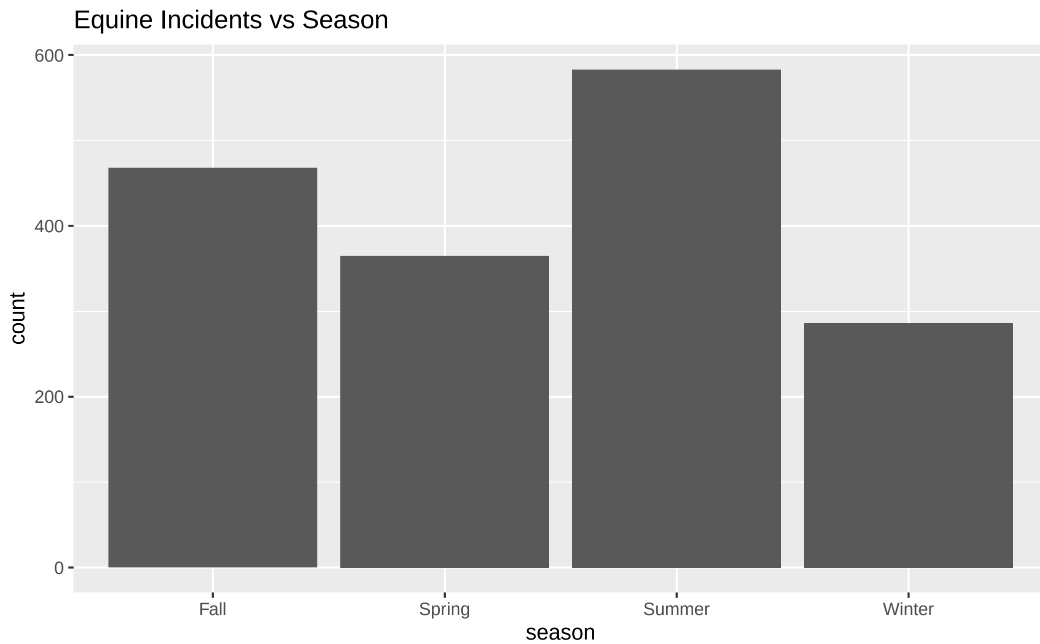
```
## `geom_smooth()` using formula = 'y ~ x'
```



- These graphs help connect to our question by showing how incident types changed over time. If certain years had higher numbers of specific incident categories, those changes could later be compared with weather patterns from those same years. This helps us look for possible trends before testing whether weather-related variables are connected to equine deaths or breakdowns. The graph does not prove that weather caused the incidents, but it helps identify time-based patterns that may be useful for the rest of the project.

iv.

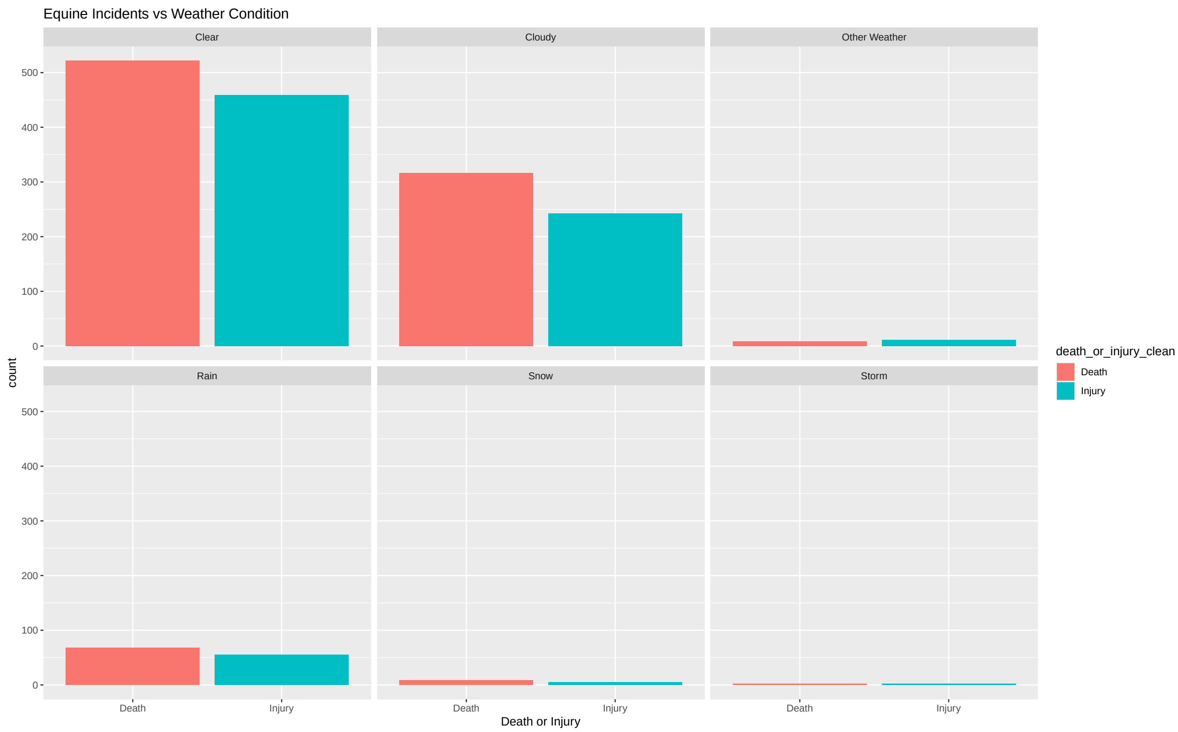
```
horses_clean %>%
  ggplot(aes(x = season)) +
  geom_bar() +
  labs(
    title = "Equine Incidents vs Season"
  )
```



- We assume that Winter has the most inclement weather out of all the seasons and yet we see that Winter has the least amount of equine deaths and injuries which could indicate a weak correlation between season and equine incidences if we take our assumption as the truth. Summer season has the most incidents but that could be due to there being more races compared to other seasons.

vi.

```
horses_clean %>%
ggplot(aes(x = death_or_injury_clean, fill = death_or_injury_clean)) +
  geom_bar() +
  facet_wrap(~ weather_group) +
  labs(
    x = "Death or Injury",
    title = "Equine Incidents vs Weather Condition")
```



- From observing the graph there does not appear to be a clear correlation between inclement weather condition and equine death or injury. The weather category with the most equine deaths and incidents is the clear weather with cloudy weather being the second most. If there was a strong correlation between inclement weather condition and equine we would expect to see rain, snow, storm, and other weather categories have higher counts compared to clear.

5. Visualization Quality and Storytelling

- For graph i. We used a scatter plot because it was the most effective way to visualize a continuous time variable and a numerical count of incidents. The scatterplot emphasizes the trend and rate of change over the time period. The individual points help the viewer recognize the specific data and the connecting line helps recognize the overall trend, the peak, the low, and the inconsistencies.
- For the incident type section, we used a bar graph first because we are looking at incident type which is categorical.This graph makes sense because it lets us compare how many incidents happened in each incident type, and we flipped the graph so the incident type names are easier to read instead of crowded on the x-axis. For the second graph, we used a line graph because we are looking at incident type by year, which helps show how the incident categories changed over time instead of only showing the total number of incidents.
- For graph iv. we used a bar graph because we were comparing two categorical variables to each other and we wanted to see which season had the most equine incidents. The x-axis is the seasons variable and the y-axis is the amount of incidents. To make the plot more visible we needed to adjust some of the parameters.

- For graph vi. we used faceted bar graphs because we were comparing multiple different categorical categories with equine incidents to see which category had the most incidents. The different colors help visualize death versus injury with respect to the specific weather condition. For visibility we also had to make the graph bigger.

6. Modeling Approach

For our modeling approach, we followed the plan from our proposal by using ideas from the blood pressure assignment and the car prices assignment. In the blood pressure assignment, we looked at relationships between variables and used linear models to understand trends. In the car prices assignment, we used a model with more than one explanatory variable. For our project, we used the number of incidents as the response variable because incident count is numeric and works with a linear model. The explanatory variables we focused on were year, season, weather condition, and incident type. These variables match the graphs we made because we looked at incidents over time, incident categories, incidents by season, and incidents by weather condition.

For our baseline model, we would use a simple heuristic, which is a simple rule for making a prediction. Our heuristic would predict the average number of incidents every time without using year, season, weather condition, or incident type. This gives us a simple comparison for our multi-variable linear model. If the multi-variable model makes better predictions than just using the average incident count, then that would suggest that the variables we chose are useful for explaining patterns in the data.

This connects to our research question because we are trying to see whether weather-related variables are connected to equine deaths and breakdowns. The incidents-over-time graph shows that incident counts changed by year, so year is an important variable to include. The incident type graphs show which categories happened the most and whether those categories changed over time. The season and weather condition graphs connect more directly to the weather part of our project because they show how incidents are spread across different seasons and weather conditions. Using a multi-variable linear model helps us look at these variables together instead of only looking at one graph or one factor at a time.

7. Model Implementation & Evaluation

```r
weather_model_data <- horses_clean %>%
  count(weather_group, incident_type, name = "incident_count", sort = TRUE)
```

```r
weather_incident_model <- lm(
  incident_count ~ weather_group + incident_type,
  data = weather_model_data
  )
```

```r
weather_incident_model %>%
  tidy()
```

term	estimate	std.error	statistic	p.value
(Intercept)	60.2019363	63.96869	0.9411156	0.3544196
weather_groupCloudy	-36.9276820	34.72112	-1.0635510	0.2963084
weather_groupOther Weather	-117.3085554	42.35578	-2.7695998	0.0096860
weather_groupRain	-81.4761905	34.82806	-2.3393838	0.0264117
weather_groupSnow	-137.5161113	49.75842	-2.7636751	0.0098256
weather_groupStorm	-148.9353715	55.75460	-2.6712662	0.0122624
incident_typeACCIDENT - ON TRACK	-17.4006454	73.53722	-0.2366236	0.8146122
incident_typeACCIDENT - TAGGED SULKY	0.7261707	70.46258	0.0103058	0.9918479
incident_typeBREAKDOWN	13.7954977	75.97072	0.1815897	0.8571680
incident_typeDRIVER/JOCKEY INJURED	-59.2019363	102.27517	-0.5788496	0.5671598
incident_typeEQUINE DEATH	185.8253822	67.59817	2.7489705	0.0101804
incident_typeEQUINE DEATH - INFECTIOUS DISEASE	-39.2380952	81.67903	-0.4803937	0.6345516
incident_typeFALL OF HORSE	11.3828748	71.04501	0.1602206	0.8738181
incident_typeFALL OF RIDER	-13.4006454	73.53722	-0.1822294	0.8566706
incident_typeRACING INCIDENT	50.6437716	68.73953	0.7367489	0.4671939
incident_typeRACING INJURY	72.9920489	67.59817	1.0797932	0.2891332
incident_typeSTEWARDS/VETS LIST	41.9761707	70.46258	0.5957228	0.5559816
incident_typeTRAINING INCIDENT	10.4761707	70.46258	0.1486771	0.8828375

```
weather_incident_model %>%
glance() %>%
select(r.squared)
```

r.squared
0.5427577

```
weather_model_data <- weather_model_data %>%
  add_predictions(weather_incident_model) %>%
    add_residuals(weather_incident_model)
```

```
weather_model_data
```

weather_group	incident_type	incident_count	pred	resid
Clear	EQUINE DEATH	539	246.027319	292.972682
Cloudy	EQUINE DEATH	323	209.099636	113.900364
Clear	RACING INJURY	163	133.193985	29.806015
Clear	STEWARDS/VETS LIST	109	102.178107	6.821893

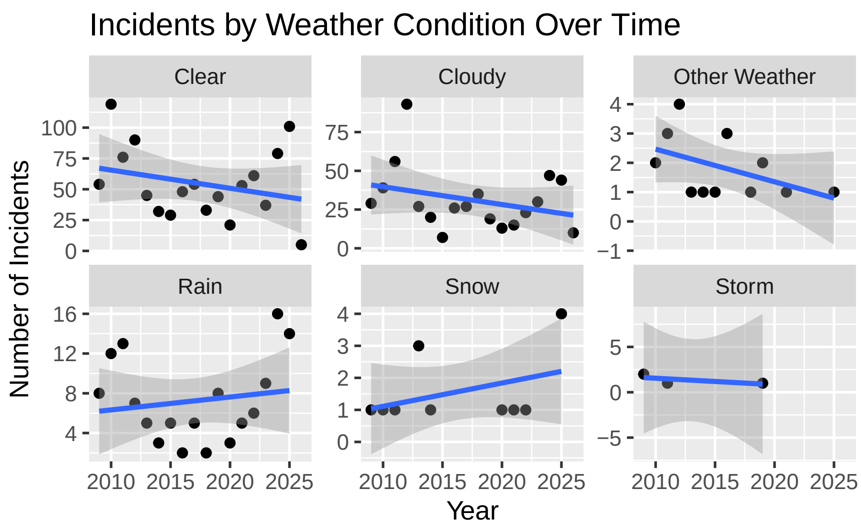
weather_group	incident_type	incident_count	pred	resid
Clear	RACING INCIDENT	104	110.845708	-6.845708
Cloudy	RACING INJURY	88	96.266303	-8.266303
Rain	EQUINE DEATH	71	164.551128	-93.551128
Cloudy	RACING INCIDENT	57	73.918026	-16.918026
Cloudy	STEWARDS/VETS LIST	54	65.250425	-11.250425
Clear	TRAINING INCIDENT	24	70.678107	-46.678107
Cloudy	TRAINING INCIDENT	21	33.750425	-12.750425
Rain	RACING INJURY	21	51.717795	-30.717795
Clear	FALL OF RIDER	15	46.801291	-31.801291
Rain	RACING INCIDENT	15	29.369517	-14.369517
Clear	FALL OF HORSE	12	71.584811	-59.584811
Other Weather	EQUINE DEATH	10	128.718763	-118.718763
Snow	EQUINE DEATH	9	108.511207	-99.511207
Rain	STEWARDS/VETS LIST	7	20.701916	-13.701916
Clear	ACCIDENT - ON TRACK	6	42.801291	-36.801291
Cloudy	FALL OF RIDER	6	9.873609	-3.873609
Clear	ACCIDENT - TAGGED SULKY	4	60.928107	-56.928107
Cloudy	FALL OF HORSE	4	34.657129	-30.657129
Clear	EQUINE DEATH - INFECTIOUS DISEASE	3	20.963841	-17.963841
Other Weather	RACING INJURY	3	15.885430	-12.885430
Other Weather	STEWARDS/VETS LIST	3	-15.130448	18.130448
Snow	RACING INCIDENT	3	-26.670403	29.670403
Cloudy	ACCIDENT - ON TRACK	2	5.873609	-3.873609
Cloudy	ACCIDENT - TAGGED SULKY	2	24.000425	-22.000425
Cloudy	EQUINE DEATH - INFECTIOUS DISEASE	2	-15.963841	17.963841
Other Weather	RACING INCIDENT	2	-6.462847	8.462848
Rain	ACCIDENT - ON TRACK	2	-38.674900	40.674900
Rain	FALL OF HORSE	2	-9.891379	11.891379
Storm	EQUINE DEATH	2	97.091947	-95.091947
Clear	BREAKDOWN	1	73.997434	-72.997434
Clear	DRIVER/JOCKEY INJURED	1	1.000000	0.000000
Cloudy	ACCIDENT - IN STARTING GATE	1	23.274254	-22.274254
Other Weather	ACCIDENT - TAGGED SULKY	1	-56.380448	57.380448
Other Weather	TRAINING INCIDENT	1	-46.630448	47.630448
Rain	ACCIDENT - IN STARTING GATE	1	-21.274254	22.274254
Rain	ACCIDENT - TAGGED SULKY	1	-20.548084	21.548084

weather_group	incident_type	incident_count	pred	resid
Rain	BREAKDOWN	1	-7.478757	8.478756
Rain	FALL OF RIDER	1	-34.674900	35.674900
Rain	TRAINING INCIDENT	1	-10.798084	11.798084
Snow	BREAKDOWN	1	-63.518677	64.518677
Snow	RACING INJURY	1	-4.322126	5.322126
Storm	FALL OF HORSE	1	-77.350561	78.350561
Storm	RACING INJURY	1	-15.741386	16.741386

- For the model implementation, we first grouped the data by weather group and incident type. Since the horse dataset records one incident per row, we used count() to create a numeric response variable called incident_count. This is different from Assignment 6, where mutate() was used because the dataset already had numeric variables and only needed a new calculated column. In our project, we needed count() because there was no total incident count column already in the data. We also used sort = TRUE to show the largest counts first, which makes the table easier to read but does not change the model. Then we used a linear model to see whether the weather group and incident type helped explain the number of incidents. tidy() shows the model estimates, glance() gives the R-squared value, and predictions and residuals help compare the model's predicted counts to the actual counts.

```
horses_clean %>%
  filter(weather_group %in% c(
    "Clear",
    "Cloudy",
    "Rain",
    "Snow",
    "Storm",
    "Other Weather"
  )) %>%
  count(year, weather_group) %>%
  ggplot(aes(x = year, y = n)) +
  geom_point() +
  geom_smooth(method = "lm") +
  facet_wrap(~ weather_group, scales = "free_y") +
  labs(
    title = "Incidents by Weather Condition Over Time",
    x = "Year",
    y = "Number of Incidents"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



- We used this graph method because our research question focuses on weather and equine incidents, but weather is a categorical variable. Since the dataset records one incident per row, we first had to count how many incidents happened in each weather condition by year. This let us compare the number of incidents over time for each weather group. We used separate panels for each weather condition so the graph would not be too crowded and so each weather pattern could be seen more clearly.

- https://dplyr.tidyverse.org/

- For the model implementation, our main focus was whether weather conditions were connected to a higher number of equine incidents. The response variable was the count of incidents, because we wanted to look at how many incidents happened under different weather conditions. The main feature was weather condition, or weather type, because that connects directly to our research question. We also included year, season, and incident type because they help give more context to the weather patterns. Year helps show whether incidents changed over time, season helps connect weather to different parts of the year, and incident type helps show whether certain kinds of incidents happened more often under certain weather conditions.

- The graphs support this because they show that incident counts were different across weather conditions and seasons. Clear and cloudy weather had higher recorded incident counts, but we have to be careful with that interpretation. A higher count does not automatically mean that clear or cloudy weather caused more incidents. It could also mean that more races happened during those weather conditions, which would create more chances for incidents to occur. This is why we looked at weather along with other variables instead of only looking at one graph by itself.

- The model helped us explore whether weather conditions were related to incident count, but we realized that our data also has important categorical variables, such as incident type and death or injury. Because of that, logistic regression may have been a better model if we were trying to predict a category like death or injury. Since our main question was focused on whether weather is connected to a higher number of incidents, we kept the linear model as an exploratory model for incident counts. However, we recognize that the model is limited, and a future version of the project could use logistic regression or a stronger weather-focused model to better test the relationship between weather and equine incidents. improvement for future work.

```r
horses_binary <- horses_clean %>%
  filter(death_or_injury_clean %in% c("Death", "Injury")) %>%
  mutate(
    death_binary = if_else(death_or_injury_clean == "Death", 1, 0)
  )

weather_death_model <- glm(
  death_binary ~ weather_group + incident_type,
  data = horses_binary,
  family = binomial
)

weather_death_model %>%
  tidy()
```

term	estimate	std.error	statistic	p.value
(Intercept)	-19.7097938	7.579521e+03	-0.0026004	0.9979252
weather_groupCloudy	0.3547858	3.804051e-01	0.9326526	0.3509994
weather_groupOther Weather	-0.9937056	1.062820e+00	-0.9349708	0.3498033
weather_groupRain	-0.1072121	6.012374e-01	-0.1783190	0.8584724
weather_groupSnow	2.0651526	3.315933e+00	0.6227968	0.5334181
weather_groupStorm	1.2972275	4.766082e+00	0.2721790	0.7854844
incident_typeACCIDENT - ON TRACK	0.0842505	8.305138e+03	0.0000101	0.9999919
incident_typeACCIDENT - TAGGED SULKY	0.1455927	8.465490e+03	0.0000172	0.9999863
incident_typeBREAKDOWN	-0.9407354	9.612932e+03	-0.0000979	0.9999219
incident_typeDRIVER/JOCKEY INJURED	0.1437253	1.315667e+04	0.0000109	0.9999913
incident_typeEQUINE DEATH	22.8866364	7.579521e+03	0.0030195	0.9975908
incident_typeEQUINE DEATH - INFECTIOUS DISEASE	39.1450857	8.972015e+03	0.0043630	0.9965188
incident_typeFALL OF HORSE	-0.0383582	7.964927e+03	-0.0000048	0.9999962
incident_typeFALL OF RIDER	0.0414382	7.917564e+03	0.0000052	0.9999958
incident_typeRACING INCIDENT	-0.0458466	7.620733e+03	-0.0000060	0.9999952
incident_typeRACING INJURY	14.6431096	7.579521e+03	0.0019319	0.9984585
incident_typeSTEWARDS/VETS LIST	14.4510813	7.579521e+03	0.0019066	0.9984788

term	estimate	std.error	statistic	p.value
incident_typeTRAINING INCIDENT	-0.0108524	7.739117e+03	-0.0000014	0.9999989

- This model is based on the logistic regression structure from Assignment 11, especially Exercises 6 and 8. Like Assignment 11, we created a binary outcome variable and used glm() with family = binomial. Instead of predicting Titanic survival, our model predicts whether an equine incident resulted in death or injury using weather group and incident type as predictors.

8. Conclusions & Recommendations

- In conclusion, our project helped us see that equine deaths and breakdowns are affected by more than one factor. We looked at incident types, time trends, seasons, and weather conditions to see what patterns showed up in the data. The graphs showed that some incident categories happened more often than others, and the number of incidents was not the same every year or every season. Even though weather was the main focus of our research question, the results show that we have to be careful when interpreting the graphs. Higher incident counts in one weather condition do not automatically mean that weather caused more incidents. It may also depend on a vast number of reasons due to the various parts of racing.
- Overall, our findings showed that equine incidents were not spread evenly across the dataset. Some incident types happened much more often than others, especially injury-related incidents, and the number of incidents changed over time. We also saw differences between seasons and weather conditions, with higher incident counts appearing in clear and cloudy weather. However, this does not prove that those weather conditions caused more incidents because there may have been more races during those conditions. Based on our graphs, weather may be connected to incident patterns, but other factors like year, season, and incident type also seem important.
- Some limitations of our research could be the phenomena known as "surivivorship bias" similar to lab 11 we did in CDS 102. The apparent relationship we are seeing between equine deaths and inclement weather is affected by the fact we don't get data on how many races were canceled due to inclement weather being too bad to race. So we can't say that weather has no correlation to equine incidents because we have to consider survivorship bias and how many horses could have been injured if they did race. If we wanted better representation we would need to add those canceled races to the dataset and observe the results.
- A useful extension for our project would be to expand our data and add more detailed information about the horses themselves. For example, we could determine the age of the horse as well as the condition to determine if the horse had any previous injuries that would lower their chances at survival. Another important extension would be to include data on if races were cancelled due to extreme weather, as we cannot see how many deaths would have been caused by weather if the races were actually done.

9. Code Quality & Reproducibility

- Someone can reproduce our results by opening the project in RStudio and running the final_report_template.Rmd file from top to bottom. The horses.csv file needs to be in the project folder so the dataset loads correctly. The required packages are tidyverse, readr, ggplot2, broom, modelr, dplyer, lubridate, and reader. Our code is organized in the same order as the report: data loading, data cleaning, EDA graphs, modeling, and evaluation. The cleaned dataset horses_clean is used for the graphs and model, so the cleaning code has to run first.
- GitHub repository

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] modelr_0.1.11   broom_1.0.12    forcats_1.0.1   purrr_1.2.1
##  [5] readr_2.1.6     tidyr_1.3.2     tibble_3.3.1    tidyverse_2.0.0
##  [9] ggplot2_4.0.2   stringr_1.6.0   lubridate_1.9.5 dplyr_1.2.0
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     lattice_0.22-7     stringi_1.8.7      hms_1.1.4
##  [5] digest_0.6.39      magrittr_2.0.4     evaluate_1.0.5     grid_4.5.2
##  [9] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-4
## [13] backports_1.5.0    mgcv_1.9-3         scales_1.4.0       cli_3.6.5
## [17] rlang_1.1.7        crayon_1.5.3       splines_4.5.2      bit64_4.6.0-1
## [21] withr_3.0.2        yaml_2.3.12        otel_0.2.0         tools_4.5.2
## [25] parallel_4.5.2     tzdb_0.5.0         vctrs_0.7.1        R6_2.6.1
## [29] lifecycle_1.0.5    bit_4.6.0          vroom_1.7.0        pkgconfig_2.0.3
## [33] pillar_1.11.1      gtable_0.3.6       glue_1.8.0         xfun_0.56
## [37] tidyselect_1.2.1   rstudioapi_0.18.0  knitr_1.51         farver_2.1.2
```

```
## [41] nlme_3.1-168       htmltools_0.5.9    rmarkdown_2.30     labeling_0.4.3
## [45] compiler_4.5.2     S7_0.2.1
```

10. References

- Equine Death and Breadown Dataset
- CDS Textbook
- Equine Breakdown, Death, Injury and Incident Database Overview
- NYS Gaming Commision Equine Breakdown
- Dates to seasons column tutorial